Yiğit Sarp Avcı

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EDUCATION

Boğaziçi University — B.Sc. in Computer Engineering 09/2023 – Expected 06/2028

- Extracurriculars: Executive Board Member & DJ at Radyo Boğaziçi; Regional Organizational Team Member at ESTIEM; Work and Travel USA participant; Student-Athlete (sailing, skiing & windsurfing).
- Volunteering: Volunteering Division Member at Industrial Engineering Club (BÜYAK); educational mentor to high school students.

Kabataş Boys' High School 09/2018 – 06/2023

- GPA: 3.78/4.00
- Extracurriculars: Debate, Model United Nations, Entrepreneurship, and Coding Clubs.
- Volunteering: Community Involvement Projects Club and TOÇEV (Tüvana Okuma İstekli Çocuk Eğitim Vakfı).

EXPERIENCE

DefineX — Backend Developer Intern 09/2026 – Present

- Developing internal data processing platforms and Model Context Protocol (MCP) integrations, streamlining operational workflows for corporate teams by automating enterprise data extraction and RESTful backend pipelines.
- Leveraging modern AI-assisted engineering workflows to accelerate enterprise backend feature delivery, implementing robust data validation schemas and optimizing API request throughput.

Akbank — Site Reliability Engineering Intern 07/2026 – 09/2026

- Maintained high availability for mission-critical core banking infrastructure by designing real-time observability dashboards and proactive alerting thresholds in Dynatrace and Grafana.
- Engaged in Linux systems engineering, server health diagnostics, and log analysis to accelerate incident detection and root-cause resolution across high-throughput production nodes.

Smash Mate — Software Engineering Intern 06/2026 – 07/2026

- Engineered player matchmaking, search discovery, and scoring microservices using Java (Quarkus), designing high-performance RESTful endpoints and core game-matching domain logic.
- Developed responsive mobile interfaces in React Native and authored comprehensive unit and integration test suites, delivering a fully functional, zero-defect pre-launch release.

TECHNICAL PROJECTS

- **Algorithmic Trading Simulator** — Real-time simulator executing Z-Score mean reversion with volatility sizing.
- **Financial Analytics Dashboard** — Platform featuring a continuous limit-order matching engine & risk models.
- **Restaurant Floor Plan SaaS** — Multi-tenant platform with interactive floor plan builder, table merging & queues.
- **Enterprise Admin Dashboard** — Internal console with role-based access control, server metrics & user workflows.
- **Full-Stack FAQ Module** — Knowledge base with SWR optimistic UI updates, database indexing, RBAC & API tests.
- **Linux IPC Game Subsystem** — 60 FPS architecture with non-blocking FIFOs, POSIX signals, dlopen plugins & buffers.
- **x86-64 CPU Scheduler** — Zero-libc simulator in assembly using raw Linux syscalls (FCFS, SJF, SRTF, PF, RR).
- **Bare-Metal RISC-V Simulator** — RV32I flight pathfinding on Spike with custom linker memory mapping & greedy routing.
- **CoreMark Performance Profiling** — Architectural study evaluating throughput across compiler flags, ILP & ISAs.
- **Unix Command Interpreter** — C-based shell featuring a custom backtracking recursive descent parser & modular state.
- **Functional Language Interpreter** — Custom engine supporting first-class closures, currying, AST evaluation & dual scoping.
- **Graph Network Analyzer** — Network analysis tool identifying articulation points & Tarjan's SCCs in $O(V + E)$ time.
- **Marketplace Matching Simulation** — Platform leveraging custom max-heaps for $O(\log N)$ priority candidate allocation.
- **React Native Minesweeper** — Mobile game featuring iterative BFS flood-fill, safe first-click, chording & haptics.
- **2048 WebAssembly Game** — C++/Qt6 implementation featuring directional grid normalization & infinite undo history.
- **State-Machine Card Game** — Terminal survival game driven by a deterministic state machine & event queues.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C, C++, TypeScript, JavaScript, SQL, x86-64 Assembly, RISC-V Assembly
- **Technologies & Frameworks:** Next.js, React Native, Quarkus, FastAPI, Express, Tailwind CSS, Prisma, PostgreSQL, Redis, Docker, WebSockets, WebAssembly, Qt6
- **Developer Tools:** Git/GitHub, Linux/POSIX, AI-Assisted Development
- **Languages:** Turkish (Native), English (Advanced), German (Beginner)
- **Hobbies:** Wakeboarding, Kiteboarding, Surfing, Bouldering, Running, Fitness, Traveling